

Security and Privacy in Federated Learning

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Agenda

- Secure Aggregation
 - Motivation
 - Threat Models
 - Protocol Examples
- Robust Aggregation
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 - Threats
 - Robust Aggregation Rules
 - Trust and Verification
- Privacy-Robustness-Utility Tradeoff

Secure Aggregation

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Even if the datasets stay with the clients, shared gradients can disclose information and enable inference attacks for both centralized and distributed FL.

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- **Malicious + (limited) access to memory**

The malicious participant additionally has access to some of the other participant's local memory (i.e. learns their secrets)

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Threat Models

- **Honest-but-curious**
Follow protocol but maximize learning
- **Malicious**
Can break the protocol
- **Malicious + (limited) access to memory**
+ Partial access to secrets

Challenges

- **User dropout**
- **Limited bandwidth and/or computation power**
- **Transmission of high dimensional information vectors (gradients)**
- **Unauthenticated network model**

Secure Aggregation Protocol by Bonawitz et al. [3]

Protocol Summary

- Uses masking vectors to encrypt individual gradients that cancel out in the aggregation
- Robust to user dropout by using secret sharing and a second masking vector
- Secure against malicious adversaries with limited memory access by using OneTime Pads and a sufficiently high threshold for secret reconstruction

Costs

for n users and vector dimension k

computation

User $O(n^2 + kn)$

Server $O(kn^2)$

communication

User $O(n + k)$

Server $O(n^2 + kn)$

storage

User $O(n + k)$

Server $O(n^2 + k)$

2 Main Variants:

■ Fully Homomorphic Encryption (FHE)

Uses additively homomorphic encryption scheme

1. Clients send encrypted gradients to the server
2. Server aggregates encrypted gradients and shares result
3. Clients jointly decrypt the updated model

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+ Privacy preservation against malicious servers

! Assumes honest clients

High communication costs

MPC/2PC variant assumes non-colluding servers

Motivation

The aggregation protocols should be robust against malicious clients trying to manipulate the training process.

Main Threats

- **Poisoning Attacks**
 - **Backdoor:** Implement sensitivity to a specific trigger
 - **Untargeted:** Degrade overall model performance
 - **Targeted:** Degrade performance on specific class(es)

Defense Mechanisms

- **Robust aggregation rules**
 - Krum
 - Trimmed Mean
 - FLTrust
- **Anomaly detection on client updates**

Robust Aggregation Rules

- **Krum [2]:** Only consider updates that are closest to other updates
 - ! Assumes few malicious clients and that honest updates cluster together → fails for non-IID data
- **Trimmed Mean [7]:** For each coordinate of the update vector, cut the top and bottom $n\%$ before averaging
 - ! neglects correlations between dimensions
for fixed, known thresholds, malicious updates can be crafted accordingly
- **FLTrust [5]:** Maintain small root dataset on server to calculate update anchor, weigh client updates by similarity to anchor
 - ! Assumes trusted, representative dataset available

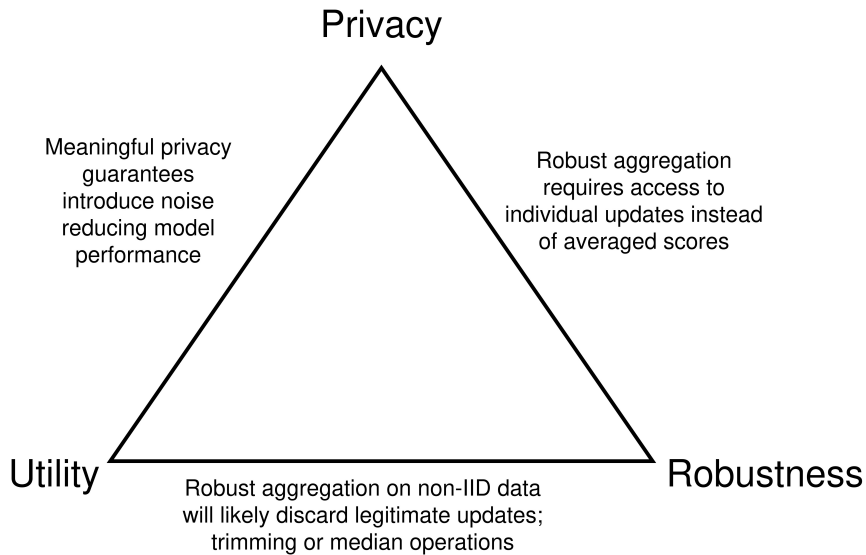
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Under highly non-IID data, honest clients naturally produce very different updates, making it hard to distinguish them from dishonest ones

- **Zero-Knowledge-Proofs** to prove that updates satisfy certain requirements [4]
 - ! Computationally expensive for large updates
- **Verifiable Computation:** Client proves it ran the training algorithm correctly, not just that the output has certain properties
 - ! Proof generation cost grows exponentially with number of gradient steps
- **Auditing/Attestation Frameworks:** audit logs + secure hardware attestation instead of cryptographic proofs
 - ! Requires legal accountability

Privacy-Robustness-Utility Tradeoff [1]



References

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1. Users pairwise agree on input perturbation pair $p_{u,v}, p_{v,u}$
2. Each user masks their gradient using the sum of all perturbations known and sends the result to the Server
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- + information theoretically secure through OneTime Pads, even for malicious adversaries with limited memory access
 - ! strong assumptions:
 - pairwise secure channels with sufficient bandwidth for secret exchange
 - protocol completion by all users

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0. Each user creates a key pair and broadcasts their public key to all other users
1. Perturbation vector agreement
 - 1.1 Secret sharing: Users distribute perturbation vector among all other users with reconstruction threshold t
2. Masking + sending of gradients
 - 2.1 Only use perturbations shared between surviving users. Reveal secrets of dropped out users if at least t users survived
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- + Robustness to user dropout
 - ! Secret exposure in case of malicious server or delayed update sending; increased communication cost

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Reducing Complexity

Share seeds for PRG instead of full vectors using DH Key Exchange via server

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Getting rid of secured channel assumption

Setup secured channels via server-mediated DH $\rightarrow$ adds PFS